

Charged-Lepton Koide Geometry from a Green-Dressed Compact Family Cycle

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Abstract

Koide's charged-lepton relation suggests that $(\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ is the natural family vector. We construct an effective compact-cycle model in which this vector is sampled from one real amplitude $Z(\phi)$ on an internal circle, while the masses are quadratic overlaps, $m_a \propto |Z(2\pi a/3)|^2$. The amplitude is built from the two lowest antiperiodic modes on the circle; their symmetric square is periodic and gives the minimal three-harmonic family space $e^{i\phi}, 1, e^{-i\phi}$. A reality condition together with the requirement that the amplitude comes from the square of one two-component spinor fixes the relative weights required by Koide's 45° geometry. The remaining orientation angle is fixed by matching one C_3 family shift to transport on the full circle: integrating out the higher Fourier harmonics gives the Berry dressing that enters the determinant term and selects $\theta_\ell = -2/9$. Using m_e and m_μ as inputs, the model predicts $m_\tau = 1776.97$ MeV.

I. INTRODUCTION

The masses of the charged leptons are accurately measured, but their hierarchy is not explained within the Standard Model. In conventional notation the charged-lepton Yukawa interaction is

$$\mathcal{L}_{Y,e} = -\bar{L}_i(Y_e)_{ij}HE_j + \text{h.c.}, \quad (1)$$

where L_i is a left-handed lepton doublet of family i , E_j is a right-handed charged-lepton singlet of family j , H is the Higgs doublet, and Y_e is a dimensionless 3×3 matrix in family space. After electroweak symmetry breaking, in unitary gauge, $H \rightarrow (0, (v + h)/\sqrt{2})^T$, with $v \simeq 246$ GeV the Higgs vacuum expectation value and h the physical Higgs field. The charged-lepton mass matrix is then $M_e = vY_e/\sqrt{2}$ [1–7]. The Higgs mechanism supplies the electroweak scale, while the eigenvalues of Y_e remain empirical inputs.

Koide observed that the charged-lepton pole masses approximately obey [8–11]

$$Q_\ell \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} \simeq \frac{2}{3}. \quad (2)$$

The striking feature of this relation is the appearance of square roots of masses. Foot gave the simplest geometric interpretation [12]: if

$$\mathbf{z} = (\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau}), \quad \hat{\mathbf{d}} = \frac{1}{\sqrt{3}}(1, 1, 1), \quad (3)$$

then Eq. (2) says that the angle between $\mathbf{z}$ and the democratic direction $\hat{\mathbf{d}}$ is $\pi/4$. A large literature has explored Koide-like relations, family-Higgs potentials, family-gauge symmetries, and possible extensions to neutrino and quark sectors [10, 13–20]. These works suggest that the $\pi/4$ angle to the democratic direction is geometrically robust, whereas fixing the vector’s position around that direction on the Koide cone is the harder problem.

This paper develops a compact-cycle effective theory for the charged-lepton sector, leaving quark masses, neutrino masses, and mixing matrices outside its present scope. Its purpose is to show how a Koide-type vector of square-root masses can arise from a single compact internal amplitude, and how the remaining angle on the Koide cone can be selected by a Green function on the internal circle.

The starting point is the pair of lowest antiperiodic modes on an internal circle, $\xi_\pm(\varphi) = e^{\pm i\varphi/2}$. Although each mode changes sign after one turn, their symmetric products are periodic and span $\{e^{i\varphi}, 1, e^{-i\varphi}\}$, a minimal three-component family space. A neutral symmetric

coefficient matrix weights these products and defines an internal family amplitude $Z(\varphi)$. A reality condition and the requirement that this matrix be the square of one underlying two-component spinor give $s^2 = 2|d|^2$, which is precisely the Koide 45° cone after sampling at three equally spaced points.

The physical masses are not taken to be linear samples of Z . The ordinary charged-lepton Yukawa matrix is obtained from an overlap of left- and right-handed internal profiles, so that in the tight C_3 limit $m_a \propto |Z(2\pi a/3)|^2$.

The remaining orientation on the cone is fixed by treating one C_3 family shift as a displacement by $2\pi/3$ on the full internal circle. Before the projection to the three-state family space, the circle contains higher Fourier harmonics. Integrating these harmonics out with the inverse-Laplacian Green function dresses one elementary shift link by a Berry phase. The determinant term winds around three such links and selects $\theta_\ell = -2/9$.

The paper is organized as follows. Sections II–IV construct the compact-cycle family amplitude from the lowest antiperiodic doublet and show how its coherent symmetric square gives the Koide cone. Sections V–VI introduce the residual C_3 clock-shift structure and compute the Green-function dressing of one elementary family shift on the full internal circle. Sections VII–VIII connect this dressed compact-cycle structure to charged-lepton Yukawa eigenvalues and show how the determinant term fixes the remaining orientation on the Koide cone. Sections IX and X state the pole-mass interpretation and give the numerical tests. The appendices discuss small departures from the self-dual cone and the stability constraints from higher Fourier harmonics.

II. ANTIPERIODIC SIGN DOUBLET AND SYMMETRIC SQUARE

We start with an internal circle. Its coordinate is a dimensionless angle

$$\varphi \sim \varphi + 2\pi. \tag{4}$$

The relevant low-lying sector is antiperiodic:

$$\psi(\varphi + 2\pi) = -\psi(\varphi). \tag{5}$$

This is the internal analogue of choosing the antiperiodic spin structure on a circle. Its Fourier modes carry half-integer internal momentum. The two lowest modes, with momenta

$\pm 1/2$, are

$$\xi_+(\varphi) = e^{+i\varphi/2}, \quad \xi_-(\varphi) = e^{-i\varphi/2}. \quad (6)$$

Both change sign under one circuit of the internal circle. They span a two-dimensional internal sign space,

$$\mathcal{H}_{\text{sign}}^{\text{AP}} = \text{span}\{\xi_+, \xi_-\}. \quad (7)$$

The word “sign” refers to the two nearest branches of the antiperiodic internal spectrum. It does not denote electric charge, and it is not by itself the particle–antiparticle operation.

The central observation is that the symmetric square of an antiperiodic doublet is periodic. Explicitly,

$$\xi_+^2 = e^{+i\varphi}, \quad (8)$$

$$\sqrt{2} \xi_+ \xi_- = \sqrt{2}, \quad (9)$$

$$\xi_-^2 = e^{-i\varphi}. \quad (10)$$

Thus

$$\text{Sym}^2 \mathcal{H}_{\text{sign}}^{\text{AP}} = \text{span}\{e^{+i\varphi}, 1, e^{-i\varphi}\}. \quad (11)$$

This gives a minimal three-component periodic carrier. In the effective theory below, this lowest symmetric-square sector is taken as the minimal three-component carrier for the charged-lepton families.

III. THE SIGN-BILINEAR ORDER PARAMETER

A family profile is encoded by a neutral symmetric sign-bilinear order parameter

$$Q_{AB} = Q_{BA}, \quad A, B \in \{+, -\}. \quad (12)$$

The field Q_{AB} is neutral with respect to Standard-Model gauge charges. It is not the Higgs doublet. It is an internal order parameter that controls the family dependence of the effective charged-lepton Yukawa eigenvalues after the ordinary Higgs field supplies the electroweak scale.

There are two equivalent ways to view Q . First, it may be a collective coordinate of the compact internal defect, specifying how the two antiperiodic sign branches are glued into a periodic family amplitude. Second, it may be introduced as a Hubbard–Stratonovich field

for a neutral sign-bilinear channel in an effective theory of the compact cycle [21, 22]. In either interpretation, the homogeneous vacuum value of Q determines a family form factor.

For a real charged-lepton mass profile it is convenient to write

$$Q = \begin{pmatrix} d & s/\sqrt{2} \\ s/\sqrt{2} & d^* \end{pmatrix}, \quad s \in \mathbb{R}, \quad d = |d|e^{i\theta}. \quad (13)$$

Here s is the singlet component. It multiplies the middle element of the symmetric square, namely the constant function. The complex number d is the oriented doublet component. It multiplies the pair of harmonics $e^{+i\varphi}$ and $e^{-i\varphi}$.

The associated family-amplitude form factor on the internal circle is

$$\begin{aligned} Z(\varphi) &= Q_{++}e^{i\varphi} + \sqrt{2}Q_{+-} + Q_{--}e^{-i\varphi} \\ &= s + de^{i\varphi} + d^*e^{-i\varphi} \\ &= s + 2|d|\cos(\varphi + \theta). \end{aligned} \quad (14)$$

The form factor $Z(\varphi)$ is dimensionless. It fixes only the family-space shape of the charged-lepton spectrum. Once the common electroweak and overlap normalization is restored, its sampled values determine the normalized square roots of the masses, while the physical masses are proportional to $|Z(\varphi_a)|^2$.

IV. COHERENT RANK ONE AND THE KOIDE CONE

A generic symmetric 2×2 matrix Q contains more freedom than is needed for the compact-cycle construction. We restrict to the coherent sector, meaning that the sign-bilinear order parameter is the symmetric square of a single two-component sign spinor q_A :

$$Q_{AB} = q_A q_B. \quad (15)$$

For a symmetric 2×2 matrix, this condition is equivalent to rank one,

$$\det Q = 0. \quad (16)$$

This is the algebraic form of coherence. For the branch-real parametrization introduced above,

$$Q = \begin{pmatrix} d & s/\sqrt{2} \\ s/\sqrt{2} & d^* \end{pmatrix}, \quad s \in \mathbb{R}, \quad d = |d|e^{i\theta}, \quad (17)$$

one has

$$\det Q = |d|^2 - \frac{s^2}{2}. \quad (18)$$

Thus the coherent rank-one condition gives

$$s^2 = 2|d|^2. \quad (19)$$

We will call this the self-dual condition.

The same condition has a more microscopic spinor-level form. Suppose the two components of the underlying spinor are related by branch reality,

$$q_- = q_+^*, \quad q_+ = r e^{i\alpha}. \quad (20)$$

Then the coherent square gives

$$Q_{++} = r^2 e^{2i\alpha} = d, \quad Q_{--} = r^2 e^{-2i\alpha} = d^*, \quad Q_{+-} = r^2 = \frac{s}{\sqrt{2}}. \quad (21)$$

Consequently,

$$|d| = r^2, \quad s = \sqrt{2} r^2, \quad (22)$$

which again gives Eq. (19). Thus branch reality is not a second independent derivation after rank one. Rather, it is the spinor-level realization of the same coherent rank-one condition, and it explains why the rank-one matrix has the real form factor

$$Z(\varphi) = s + d e^{i\varphi} + d^* e^{-i\varphi}. \quad (23)$$

In this sense the self-dual condition is not a tuned potential minimum; it is the kinematics of a coherent branch-real symmetric square.

We now connect this self-dual condition to Koide geometry. Write

$$Z(\varphi) = s [1 + \kappa \cos(\varphi + \theta)], \quad \kappa = \frac{2|d|}{s}. \quad (24)$$

Equation (19) is equivalent to

$$\kappa = \sqrt{2}. \quad (25)$$

Equivalently, in the inner product

$$\langle f, g \rangle = \int_0^{2\pi} \frac{d\varphi}{2\pi} f^*(\varphi) g(\varphi), \quad (26)$$

the singlet part of Z has norm s^2 , while the oscillating doublet part $de^{i\varphi} + d^*e^{-i\varphi}$ has norm $2|d|^2$. Thus Eq. (19) says that the singlet and doublet pieces of the compact family amplitude have equal norm.

Sampling at three equally spaced phases,

$$\varphi_a = \frac{2\pi a}{3}, \quad a = 0, 1, 2, \quad (27)$$

gives the normalized sampled amplitude

$$z_a \equiv \frac{Z(\varphi_a)}{s} = 1 + \sqrt{2} \cos\left(\theta + \frac{2\pi a}{3}\right). \quad (28)$$

On a branch where the three sampled amplitudes are positive, the vector (z_0, z_1, z_2) is proportional to the physical square-root mass vector. More generally, it is the signed square-root vector, while the physical masses remain proportional to $|Z(\varphi_a)|^2$.

Using

$$\sum_{a=0}^2 \cos\left(\theta + \frac{2\pi a}{3}\right) = 0, \quad \sum_{a=0}^2 \cos^2\left(\theta + \frac{2\pi a}{3}\right) = \frac{3}{2}, \quad (29)$$

one obtains

$$\sum_a z_a = 3, \quad \sum_a z_a^2 = 6. \quad (30)$$

Therefore, on the positive square-root branch,

$$\frac{\sum_a m_a}{\left(\sum_a \sqrt{m_a}\right)^2} = \frac{\sum_a z_a^2}{\left(\sum_a z_a\right)^2} = \frac{6}{9} = \frac{2}{3}. \quad (31)$$

Equivalently, the square-root mass vector has equal norm along the democratic axis and orthogonal to it. This is the 45° Koide cone.

V. RESIDUAL C_3 CLOCK AND SHIFT

The family space used for the charged leptons is the three-dimensional carrier

$$\mathcal{H}_f = \text{span}\{|0\rangle, |1\rangle, |2\rangle\}. \quad (32)$$

The C_3 clock operator is defined by

$$\hat{V}|a\rangle = \omega^a|a\rangle, \quad \omega = e^{2\pi i/3}. \quad (33)$$

The eigenstates of $\hat{V}$ are the charged-lepton basis states. We use the ordering

$$|0\rangle = |\tau\rangle, \quad |1\rangle = |\mu\rangle, \quad |2\rangle = |e\rangle. \quad (34)$$

The eigenvalue ω^a corresponds to the point $\varphi_a = 2\pi a/3$ on the internal circle.

The shift operator moves from one clock state to the next. The bare shift is

$$\hat{U}_0|a\rangle = |a+1\rangle, \quad |a+3\rangle = |a\rangle, \quad (35)$$

and obeys

$$\hat{U}_0\hat{V} = \omega^{-1}\hat{V}\hat{U}_0, \quad \hat{U}_0^3 = \mathbf{1}. \quad (36)$$

This is the standard finite clock-shift algebra. The important point in the present model is that $\hat{U}_0$ is not treated as an isolated discrete operation. It is the projection of a displacement by

$$\Delta = \frac{2\pi}{3} \quad (37)$$

inside the full internal circle. The full circle contains infinitely many harmonics. When the theory is reduced to the three-dimensional charged-lepton sector, the discarded harmonics can dress the effective shift operator. The next section computes this dressing.

VI. GREEN-DRESSED SHIFT FROM A COMPACT BERRY CONNECTION

A C_3 family shift is the projection of a displacement on the full internal circle. One elementary step moves a clock point by $\Delta = 2\pi/3$. Before the projection to the three-dimensional family carrier, the compact circle contains all Fourier harmonics. The nonzero harmonics can therefore dress the elementary shift operator. We describe this dressing by a compact Berry connection $a(\varphi)$ on the internal circle. This is the standard slow-fast adiabatic setting: after fast internal modes are eliminated, transport of the slow family frame is encoded by a Berry connection [23–26].

The fast compact sector is taken to have the canonical phase-field stiffness

$$K = -\partial_\varphi^2. \quad (38)$$

On zero-average functions its inverse is the circle Green function,

$$G = K^{-1}, \quad Ge^{in\varphi} = \frac{1}{n^2}e^{in\varphi}, \quad n \neq 0. \quad (39)$$

The inner product used below is

$$\langle f, g \rangle = \int_0^{2\pi} \frac{d\varphi}{2\pi} f^*(\varphi)g(\varphi). \quad (40)$$

An elementary oriented link compares the two endpoints of the displacement. We encode it by the neutral endpoint source

$$j_\Delta(\varphi) = \delta_{2\pi}(\varphi) - \delta_{2\pi}(\varphi - \Delta), \quad (41)$$

where $\delta_{2\pi}$ is normalized with respect to $d\varphi/(2\pi)$. Its Fourier components are

$$(j_\Delta)_n = 1 - e^{-in\Delta}. \quad (42)$$

The factor $1 - e^{-in\Delta}$ measures the response of the n th harmonic to one family step.

At the level of the compact-cycle effective theory, the one-link matching is specified by the linear response of the fast Berry sector to the endpoint dipole source. Keeping only the leading Gaussian response of the nonzero Fourier harmonics gives the most economical translation-invariant functional

$$S_B[a; j_\Delta] = -\kappa_B \langle a, Ka \rangle + g_B \langle j_\Delta, a \rangle. \quad (43)$$

Here $a(\varphi)$ is the compact Berry-response field, g_B is the endpoint charge of one elementary link, and κ_B fixes the quadratic normalization of the fast compact sector. The first term is the Gaussian cost for exciting nonzero circle harmonics, while the second term is the coupling of those harmonics to the two endpoints of the family shift.

The minimal compact normalization used below is

$$g_B = 1, \quad \kappa_B = \pi^2. \quad (44)$$

In this canonical sector Eq. (43) becomes

$$S_B[a; j_\Delta] = -\pi^2 \langle a, Ka \rangle + \langle j_\Delta, a \rangle. \quad (45)$$

This functional enters the real-time matching amplitude as $\exp(iS_B)$.

The Gaussian integral is evaluated by shifting the Berry-response field to its classical value in the presence of the endpoint source. The stationary configuration satisfies

$$Ka_{\text{cl}} = \frac{1}{2\pi^2} j_\Delta, \quad a_{\text{cl}} = \frac{1}{2\pi^2} G j_\Delta. \quad (46)$$

Equivalently, completing the square gives

$$-\pi^2 \langle a, Ka \rangle + \langle j, a \rangle = -\pi^2 \left\langle a - \frac{1}{2\pi^2} G j, K \left(a - \frac{1}{2\pi^2} G j \right) \right\rangle + \frac{1}{4\pi^2} \langle j, G j \rangle. \quad (47)$$

Thus the normalized one-link amplitude is

$$\frac{\int \mathcal{D}a \exp\{iS_B[a; j_\Delta]\}}{\int \mathcal{D}a \exp\{iS_B[a; 0]\}} = e^{i\gamma(\Delta)}, \quad (48)$$

with

$$\gamma(\Delta) = \frac{1}{4\pi^2} \langle j_\Delta, G j_\Delta \rangle. \quad (49)$$

Using Eq. (42), one finds

$$\langle j_\Delta, G j_\Delta \rangle = \sum_{n \neq 0} \frac{|1 - e^{-in\Delta}|^2}{n^2} = 4 \sum_{n=1}^{\infty} \frac{1 - \cos(n\Delta)}{n^2}. \quad (50)$$

Therefore

$$\gamma(\Delta) = \frac{1}{\pi^2} \sum_{n=1}^{\infty} \frac{1 - \cos(n\Delta)}{n^2}. \quad (51)$$

Restoring the general normalization of Eq. (43), the same completing-square step gives

$$\gamma(\Delta) = \frac{g_B^2}{4\kappa_B} \langle j_\Delta, G j_\Delta \rangle. \quad (52)$$

Using Eq. (50), this can be written as

$$\gamma(\Delta) = \lambda_B \frac{1}{\pi^2} \sum_{n=1}^{\infty} \frac{1 - \cos(n\Delta)}{n^2}, \quad \lambda_B = \frac{g_B^2 \pi^2}{\kappa_B}. \quad (53)$$

With the canonical compact normalization of Eq. (44), one has $\lambda_B = 1$.

For a C_3 step, $\Delta = 2\pi/3$. Harmonics divisible by three are blind to this separation, while the remaining harmonics obey

$$1 - \cos\left(\frac{2\pi n}{3}\right) = \frac{3}{2}, \quad 3 \nmid n. \quad (54)$$

Hence

$$\gamma_{C_3} = \frac{3}{2\pi^2} \sum_{3 \nmid n} \frac{1}{n^2} = \frac{3}{2\pi^2} \left(1 - \frac{1}{9}\right) \zeta(2) = \frac{3}{2\pi^2} \cdot \frac{8}{9} \cdot \frac{\pi^2}{6} = \frac{2}{9}. \quad (55)$$

This is the Green dressing of one elementary family-shift link in the minimal compact Berry sector.

The effective one-step shift operator is therefore

$$\hat{U}_{\text{eff}} = e^{i\gamma_{C_3}} \hat{U}_0. \quad (56)$$

The oppositely oriented link is represented by

$$\hat{U}_{\text{eff}}^\dagger = e^{-i\gamma_{C_3}} \hat{U}_0^\dagger. \quad (57)$$

A circuit around the three-state family triangle contains three elementary links, so

$$\hat{U}_{\text{eff}}^3 = e^{3i\gamma_{C_3}} \mathbf{1} = e^{2i/3} \mathbf{1}. \quad (58)$$

The corresponding family holonomy is

$$\delta_f = 3\gamma_{C_3} = \frac{2}{3}. \quad (59)$$

The result of this section is the matched elementary link $\hat{U}_0 \rightarrow \hat{U}_{\text{eff}}$. In the next step, the determinant term combines three such oriented links and converts the compact Berry phase into the orientation angle of the charged-lepton vector on the Koide cone.

VII. QUADRATIC YUKAWA OVERLAP

We now connect the compact-cycle family amplitude to the charged-lepton Yukawa eigenvalues. The amplitude $Z(\varphi)$ becomes an operator on the three-dimensional clock space by replacing the phase $e^{i\varphi}$ with the clock operator $\hat{V}$:

$$Z(\hat{V}) = s\mathbf{1} + d\hat{V} + d^*\hat{V}^\dagger. \quad (60)$$

Since $\hat{V}$ is diagonal in the charged-lepton basis,

$$\hat{V}|a\rangle = e^{2\pi ia/3}|a\rangle, \quad a = 0, 1, 2, \quad (61)$$

the operator $Z(\hat{V})$ samples the compact amplitude at the three clock points:

$$Z(\hat{V})|a\rangle = Z(\varphi_a)|a\rangle, \quad \varphi_a = \frac{2\pi a}{3}. \quad (62)$$

The physical masses are not assumed to be linear samples of Z . In the four-dimensional effective theory, the charged-lepton Yukawa matrix arises from an overlap of left- and right-handed internal profiles. To make this explicit, write the internal-circle extensions of the Standard-Model lepton doublet and charged-lepton singlet as

$$L(x, \varphi) = \sum_a L_a(x) f_a^L(\varphi), \quad E(x, \varphi) = \sum_b E_b(x) f_b^R(\varphi). \quad (63)$$

A local Yukawa interaction on the compact cycle has the form

$$\mathcal{L}_Y = -y_0 \int_0^{2\pi} \frac{d\varphi}{2\pi} L(x, \varphi) H(x) E(x, \varphi) + \text{h.c.} \quad (64)$$

and gives the family-space Yukawa matrix

$$(Y_\ell)_{ab} = y_0 \int_0^{2\pi} \frac{d\varphi}{2\pi} [f_a^L(\varphi)]^* f_b^R(\varphi). \quad (65)$$

In the aligned compact-family sector the left- and right-handed profiles carry the same family amplitude:

$$f_a^L(\varphi) = Z(\varphi)w_a(\varphi), \quad f_a^R(\varphi) = Z(\varphi)w_a(\varphi). \quad (66)$$

Here $w_a(\varphi)$ is a clock-localized profile centered at $\varphi_a = 2\pi a/3$. In the tight C_3 limit,

$$w_a^*(\varphi)w_b(\varphi) \simeq \delta_{ab} \delta_{2\pi}(\varphi - \varphi_a), \quad (67)$$

where $\delta_{2\pi}$ is normalized with respect to $d\varphi/(2\pi)$. The Yukawa matrix is then diagonal in the clock basis:

$$(Y_\ell)_{ab} \simeq y_0 |Z(\varphi_a)|^2 \delta_{ab}. \quad (68)$$

Equivalently,

$$Y_\ell = y_* Z^\dagger(\hat{V}) Z(\hat{V}), \quad (69)$$

up to the common normalization y_* , which includes the overlap normalization of the localized profiles.

After electroweak symmetry breaking, the charged-lepton masses are therefore

$$m_a = \frac{v}{\sqrt{2}} y_* |Z(\varphi_a)|^2. \quad (70)$$

Using the self-dual Koide-cone form

$$Z(\varphi) = s \left[1 + \sqrt{2} \cos(\varphi + \theta) \right], \quad (71)$$

one obtains the one-scale spectrum

$$m_a = M \left[1 + \sqrt{2} \cos \left(\frac{2\pi a}{3} + \theta \right) \right]^2, \quad a = 0, 1, 2, \quad (72)$$

with

$$M = \frac{v}{\sqrt{2}} y_* s^2. \quad (73)$$

At this stage the Koide cone has fixed the relative singlet and doublet weights in Z , but it has not fixed the remaining orientation angle θ on the cone. The next section shows how the Green-dressed compact shift selects this angle.

VIII. DETERMINANT ANISOTROPY

The Green-function matching of the compact Berry sector dresses one elementary family shift by a phase. In the minimal compact normalization found in Sec. VI,

$$\hat{U}_{\text{eff}} = e^{i\gamma_{C_3}} \hat{U}_0, \quad \gamma_{C_3} = \frac{2}{9}. \quad (74)$$

A full circuit around the three-state family triangle contains three such oriented links, so

$$\hat{U}_{\text{eff}}^3 = e^{i\delta_f} \mathbf{1}, \quad \delta_f = 3\gamma_{C_3} = \frac{2}{3}. \quad (75)$$

The determinant term is built from the compact-cycle family amplitude in the shift sector. To avoid confusing this object with the physical Yukawa matrix Y_ℓ , we denote it by

$$\mathcal{Z}_{\delta_f} = s\mathbf{1} + d\hat{U}_{\text{eff}} + d^*\hat{U}_{\text{eff}}^\dagger. \quad (76)$$

This is the shift-space analogue of the clock-space amplitude $Z(\hat{V})$. The clock-space amplitude gives the Yukawa eigenvalues through the quadratic overlap of Sec. VII; the shift-space amplitude controls the phase-sensitive determinant anisotropy.

Using $\hat{U}_{\text{eff}}^3 = e^{i\delta_f} \mathbf{1}$, one finds

$$\det \mathcal{Z}_{\delta_f} = s^3 - 3s|d|^2 + e^{i\delta_f} d^3 + e^{-i\delta_f} d^{*3}. \quad (77)$$

The first two terms are independent of the orientation of d . The last two terms depend on the phase

$$d = |d|e^{i\theta} \quad (78)$$

and therefore select the remaining angle on the Koide cone.

A symmetry-compatible effective potential may be written as

$$V_{\text{eff}}(Q) = V_{\text{radial}}(Q) + \lambda_{\text{SD}} |\det Q|^2 - \rho \text{Re} \det \mathcal{Z}_{\delta_f} + \dots. \quad (79)$$

Here V_{radial} stabilizes the overall magnitude of Q , while $\lambda_{\text{SD}} > 0$ penalizes departures from the coherent rank-one cone. The orientation-dependent part is

$$V_\theta = -2\rho |d|^3 \cos(3\theta + \delta_f). \quad (80)$$

For $\rho > 0$ the minimum satisfies

$$3\theta + \delta_f = 0 \quad \text{mod } 2\pi. \quad (81)$$

Thus, in the physical chamber used for the charged-lepton ordering,

$$\theta_\ell = -\frac{\delta_f}{3} = -\gamma_{C_3} = -\frac{2}{9}. \quad (82)$$

The role of the Berry calculation is therefore to fix the angle that was still free after imposing the Koide cone. The cone condition determines the shape of the square-root mass vector; the Green-dressed family shift determines its orientation on that cone. Substituting the selected value of θ into the quadratic-overlap spectrum gives

$$m_a = M \left[1 + \sqrt{2} \cos \left(\frac{2\pi a}{3} - \frac{2}{9} \right) \right]^2, \quad a = 0, 1, 2, \quad (83)$$

where the ordering is

$$|0\rangle = |\tau\rangle, \quad |1\rangle = |\mu\rangle, \quad |2\rangle = |e\rangle. \quad (84)$$

The same result can be viewed as a dressed three-link vertex. The term d^3 corresponds to the oriented process $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ around the family triangle. Each elementary link carries the Berry phase γ_{C_3} , so the determinant vertex carries the total phase $\delta_f = 3\gamma_{C_3}$. This three-link phase is what converts the compact Berry dressing into the charged-lepton orientation $\theta_\ell = -2/9$.

IX. POLE-MASS INTERPRETATION

Koide's relation is conventionally a relation among pole masses. The effective relation here is formulated in the same language. Define

$$Y_\ell^{\text{pole}} \equiv \frac{\sqrt{2}}{v} M_\ell^{\text{pole}}. \quad (85)$$

The compact-cycle matching condition is

$$Y_\ell^{\text{pole}} = y_* Z^\dagger(\hat{V}) Z(\hat{V}). \quad (86)$$

This should not be read as a statement about arbitrary running Yukawa couplings at an arbitrary renormalization scale. Pole masses include QED and electroweak self-energy effects, and running Yukawas generally do not preserve Koide-type relations without additional structure. A UV completion must therefore explain why the compact-cycle matching is reproduced at the pole-mass level, or else supply a protective mechanism for the running relation, as in Sumino-type approaches [14, 15].

X. NUMERICAL TESTS

The cleanest noncircular test uses only m_e and m_μ to determine the scale and the azimuth, and then predicts m_τ . On the self-dual cone,

$$m_a = M \left[1 + \sqrt{2} \cos \left(\frac{2\pi a}{3} + \theta \right) \right]^2. \quad (87)$$

Solving

$$\frac{m_e}{m_\mu} = \left[\frac{1 + \sqrt{2} \cos(4\pi/3 + \theta)}{1 + \sqrt{2} \cos(2\pi/3 + \theta)} \right]^2 \quad (88)$$

on the physical branch gives

$$\theta_{e\mu} = -0.222222047 \dots, \quad (89)$$

or

$$\theta_{e\mu} + \frac{2}{9} \simeq 1.75 \times 10^{-7}. \quad (90)$$

The corresponding blind tau prediction is

$$m_\tau^{\text{pred}}(m_e, m_\mu) = m_\mu \left[\frac{1 + \sqrt{2} \cos \theta_{e\mu}}{1 + \sqrt{2} \cos(2\pi/3 + \theta_{e\mu})} \right]^2 = 1776.9690 \text{ MeV}. \quad (91)$$

It is also useful to display the strict minimal Green value, $\theta = -2/9$. The dimensionless intensities are

$$|Z_\tau|^2 = 5.6617260126, \quad |Z_\mu|^2 = 0.3366458723, \quad |Z_e|^2 = 0.0016281151. \quad (92)$$

Fixing the scale from m_μ alone gives

$$M_{(\mu)} = \frac{m_\mu}{|Z_\mu|^2} = 313.8561444 \text{ MeV}, \quad (93)$$

and therefore

$$m_\tau^{\text{pred}}(m_\mu) = 1776.9675 \text{ MeV}, \quad m_e^{\text{pred}}(m_\mu) = 0.5109939 \text{ MeV}. \quad (94)$$

Fixing the scale from m_e alone gives

$$m_\tau^{\text{pred}}(m_e) = 1776.9850 \text{ MeV}. \quad (95)$$

The current pole masses are [7]

$$m_e = 0.51099895000 \text{ MeV}, \quad m_\mu = 105.6583755 \text{ MeV}, \quad m_\tau = 1776.93 \pm 0.09 \text{ MeV}. \quad (96)$$

TABLE I. Charged-lepton tests of the self-dual compact-family spectrum.

Input	Output	Value
m_e, m_μ	$\theta_{e\mu}$	$-0.222222047\dots$
m_e, m_μ	m_τ	1776.9690 MeV
$m_\mu, \theta = -2/9$	m_τ	1776.9675 MeV
$m_\mu, \theta = -2/9$	m_e	0.5109939 MeV
strict $\theta = -2/9$	$\delta(m_e/m_\mu)/(m_e/m_\mu)$	-9.83×10^{-6}
$m_e, \theta = -2/9$	m_τ	1776.9850 MeV
PDG	m_τ	1776.93 ± 0.09 MeV

The strict minimal value is not an exact fit to the best-measured ratio:

$$\left(\frac{m_e}{m_\mu}\right)_{\theta=-2/9} = 0.00483628414, \quad \left(\frac{m_e}{m_\mu}\right)_{\text{exp}} = 0.00483633169. \quad (97)$$

The relative residual is about -9.83×10^{-6} . In the effective theory this is naturally interpreted as the size of finite compact-cycle, radiative, or higher-harmonic matching corrections to the canonical minimal result.

XI. DISCUSSION

The construction separates the charged-lepton problem into two parts. The Koide cone follows from the coherent branch-real symmetric square of the lowest antiperiodic doublet: the resulting compact amplitude has equal singlet and doublet norm. The remaining orientation on that cone is fixed by the compact Berry sector. In the minimal normalization, the inverse-Laplacian response of the full internal circle dresses one elementary C_3 shift link by $\gamma_{C_3} = 2/9$, and the determinant vertex converts the three-link holonomy into the charged-lepton angle $\theta_\ell = -2/9$.

The numerical success of the construction relies on two matching assumptions. First, the compact Berry sector is taken in the canonical normalization $\lambda_B = 1$ of Eq. (53). A microscopic completion must explain why additional compact charges, massive fast modes, or a bare link holonomy do not appreciably shift this value. Second, the matching is interpreted at the pole-mass level, as in Eq. (86). This requires either a UV mechanism that reproduces

the compact-cycle relation for pole masses or a protection mechanism for the corresponding running relation.

The near-node origin of the electron mass makes higher-harmonic stability a sharp constraint. Since Z_e is small at the selected angle, even a tiny higher-harmonic correction to the compact amplitude can produce an enhanced relative shift in m_e . Appendix B shows that harmonics beyond $k = 0, \pm 1$ must therefore be symmetry-forbidden or strongly suppressed if the observed 10^{-5} -level accuracy of the charged-lepton ratios is to remain stable.

The present effective theory is restricted to charged leptons and does not address quark masses, neutrino masses, or flavor mixing. If the order-parameter sector is dynamical, fluctuations of its radial and phase modes generate corrections to charged-lepton Yukawa couplings. In the exactly aligned limit these corrections remain diagonal in the charged-lepton clock basis, while misalignment would induce lepton-flavor-violating operators constrained by searches such as $\mu^+ \rightarrow e^+ \gamma$ [27], $\tau \rightarrow 3\ell$ [28], and $\tau \rightarrow \ell \gamma$ [29].

The main theoretical task is therefore to embed the compact-cycle effective description in a microscopic antiperiodic Dirac/Yukawa model. Such a completion should reproduce the three essential low-energy ingredients identified here: a branch-real coherent symmetric square giving the Koide cone, a unit endpoint-dipole Berry source for one elementary family shift, and an inverse-Laplacian compact response with no independent bare holonomy.

Appendix A: Small deformations away from the self-dual cone

To parametrize small departures from the ideal self-dual spectrum, write

$$s^2 = 2|d|^2(1 + \eta), \quad \theta = -\frac{2}{9} + \Delta\theta. \quad (\text{A1})$$

Then

$$\beta = \frac{2|d|}{s} = \sqrt{\frac{2}{1 + \eta}} = \sqrt{2} \left(1 - \frac{\eta}{2} + O(\eta^2) \right), \quad (\text{A2})$$

and

$$m_a = Mx_a^2, \quad x_a = 1 + \beta \cos \left(\frac{2\pi a}{3} + \theta \right). \quad (\text{A3})$$

Using the trigonometric identities for three equally spaced points,

$$\sum_{a=0}^2 \cos \left(\frac{2\pi a}{3} + \theta \right) = 0, \quad \sum_{a=0}^2 \cos^2 \left(\frac{2\pi a}{3} + \theta \right) = \frac{3}{2}, \quad (\text{A4})$$

one obtains

$$Q_\ell = \frac{1}{3} + \frac{\beta^2}{6} = \frac{2}{3} - \frac{\eta}{3} + O(\eta^2). \quad (\text{A5})$$

Thus the deviation of Q_ℓ from $2/3$ measures departure from the self-dual cone.

At $\eta = 0$ and $\theta = -2/9$, the two independent logarithmic ratios respond as

$$\delta \ln \frac{m_e}{m_\mu} \simeq 23.06 \eta + 56.14 \Delta\theta, \quad (\text{A6})$$

$$\delta \ln \frac{m_\mu}{m_\tau} \simeq 1.303 \eta - 4.917 \Delta\theta, \quad (\text{A7})$$

up to terms quadratic in η and $\Delta\theta$.

Appendix B: Higher harmonic stability

The truncation

$$Z(\varphi) = s + d e^{i\varphi} + d^* e^{-i\varphi} \quad (\text{B1})$$

is important because the electron sits close to a node. Suppose higher harmonics shift the amplitude by

$$\delta Z(\varphi) = c_2 e^{2i\varphi} + c_2^* e^{-2i\varphi} + \dots \quad (\text{B2})$$

The relative electron-mass shift is approximately

$$\frac{\delta m_e}{m_e} \simeq 2 \frac{\delta Z_e}{Z_e}. \quad (\text{B3})$$

At $\theta = -2/9$, $Z_e \simeq 0.04035$. Maintaining relative accuracy at the 10^{-5} level therefore requires $|\delta Z_e| \lesssim 2 \times 10^{-7}$. Any ultraviolet completion must therefore suppress higher harmonics by symmetry or by a strong dynamical hierarchy.

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